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Embryology of the Human Genitourinary Tract

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Keywords

bladder; embryology; external genitalia; gonads; human genitourinary tract; prostate; ureters and kidneys; uterus; vagina

Questions

1. The fetal kidneys develop from which of the following embryonic structures?
    a. Paraxial (somite) mesoderm
    b. Intermediate mesoderm
    c. Neural tube
    d. Lateral mesoderm
2. At what gestational time point does the metanephros development begin?
    a. 2 weeks
    b. 3 weeks
    c. 4 weeks
    d. 5 weeks
3. Which of the following statements is TRUE of the metanephric development?
    a. It requires the reciprocal inductive interaction between Müllerian duct and metanephric mesenchyme.

b. The calyces, pelvis, and ureter derive from the differentiation of the metanephric mesenchyme.
c. Older, more differentiated nephrons are located at the periphery of the developing kidney, whereas newer, less differentiated nephrons are found near the juxtamedullary region.
d. In humans, although renal maturation continues postnatally, nephrogenesis is completed by birth.

4. The fused lower pole of the horseshoe kidney is trapped by which of the following structures during the ascent?
   a. Inferior mesenteric artery
   b. Superior mesenteric artery
   c. Celiac artery
   d. Common iliac artery
5. The homozygous gene disruption (gene knockout) in which of the following molecules does NOT lead to a significant renal maldevelopment in mice?
   a. WT-1
   b. Pax-2
   c. GDNF
   d. p53
6. Which of the following statements is NOT TRUE of GDNF?
   a. It is a ligand for the RET receptor tyrosine kinase.
   b. GDNF gene knockout mice demonstrate an abnormal renal development.
   c. It is expressed in the metanephric mesenchyme but not in the ureteric bud.
   d. GDNF arrests the ureteric bud growth in vitro.
7. The bladder trigone develops from which of the following structures?
   a. Mesonephric ducts
   b. Müllerian ducts
   c. Urogenital sinus
   d. Metanephric mesenchyme
8. The urachus involutes to become:
   a. verumontanum.

b. the median umbilical ligament.
c. appendix testicle.
d. epoophoron.

9. Which of the following statements is NOT TRUE of bladder development?
    a. The bladder body and trigone is derived from the urogenital sinus and not the terminal portion of the mesonephric ducts.
    b. Bladder compliance seems to be low during early gestation, and it gradually increases thereafter.
    c. Epithelial-mesenchymal inductive interactions appear to be necessary for proper bladder development.
    d. Histologic evidence of smooth muscle differentiation begins near the bladder neck and proceeds toward the bladder dome.
10. The primordial germ cell migration and the formation of the genital ridges begin at which time point during gestation?
    a. Third week
    b. Fifth week
    c. Seventh week
    d. Ninth week
11. Which of the following statements is NOT TRUE of the paramesonephric (müllerian) ducts?
    a. Both male and female embryos form paramesonephric (müllerian) ducts.
    b. In male embryos, the paramesonephric ducts degenerate under the influence of the MIS (müllerian-inhibiting substance) produced by the Leydig cells.
    c. In male embryos, the paramesonephric ducts become the appendix testis and the prostatic utricle.
    d. In female embryos, the paramesonephric ducts form the female reproductive tract, including fallopian tubes, uterus, and upper vagina.
12. Which of the following structures in the male reproductive tract develops from the urogenital sinus?
    a. Vas deferens
    b. Testis
    c. Prostate

d. Appendix epididymis

13. Which of the following statements is NOT TRUE of normal prostate development?
    a. It requires the conversion of testosterone into dihydrotestosterone by 5α-reductase.
    b. It is dependent on epithelial-mesenchymal interactions under the influence of androgens.
    c. It is first seen at the 10th to 12th week of gestation.
    d. It requires the effects of MIS.
14. In female embryos, the remnants of the mesonephric ducts persist as the following structures EXCEPT:
    a. epoophoron.
    b. paroophoron.
    c. hymen.
    d. Gartner duct cysts.
15. Which of the following statements is NOT TRUE of the external genitalia development?
    a. The appearance of the external genitalia is similar in male and female embryos until the 8th week.
    b. The external genital appearance of males who are deficient in 5α-reductase is similar to that of females.
    c. In males, the formation of distal glandular urethra may occur by the fusion of urethral folds proximally and the ingrowth of ectodermal cells distally.
    d. In females, the urethral folds become the labia majora, and the labioscrotal folds become the labia minora.
16. The testicles descend to the level of internal inguinal ring by which time point during gestation?
    a. Sixth week
    b. Third month
    c. Sixth month
    d. Ninth month
17. Which of the following statements is NOT TRUE of the sex-determining region of the Y chromosome (SRY)?
    a. Its expression triggers the primitive sex cord cells to differentiate into the Sertoli cells.

b. Approximately 25% of sex reversal conditions in humans are attributable to SRY mutations.
c. It is located on the short arm of the Y chromosome.
d. It causes the regression of mesonephric ducts.

Answers

1. **b. Intermediate mesoderm.** Mammals develop three kidneys in the course of intrauterine life. The embryonic kidneys are, in order of their appearance, the pronephros, the mesonephros, and the metanephros. The first two kidneys regress in utero, and the third becomes the permanent kidney. In terms of embryology, all three kidneys develop from the intermediate mesoderm.
2. **c. 4 weeks.** The definitive kidney, metanephros, forms in the sacral region as a pair of new structures, called the ureteric buds, sprouts from the distal portion of the nephric duct and comes in contact with the blastema of metanephric mesenchyme at about 4 weeks.
3. **d. In humans, although renal maturation continues postnatally, nephrogenesis is completed by birth.** It requires the inductive interaction between the ureteric bud and metanephric mesenchyme. The calyces, pelvis, and ureter derive from the ureteric bud. Older, more differentiated nephrons are located in the inner part of the kidney near the juxtamedullary region. In humans, although renal maturation continues to take place postnatally, nephrogenesis is completed before birth.
4. **a. Inferior mesenteric artery.** The inferior poles of the kidneys may fuse, forming a horseshoe kidney that crosses over the ventral side of the aorta. During ascent, the fused lower pole becomes trapped under the inferior mesenteric artery and thus does not reach its normal site.
5. **d. p53.** Mutant *WT-1* mice do not form ureteric buds, and in *Pax-2* gene knockout mice, no nephric ducts, müllerian ducts, ureteric buds, or metanephric mesenchyme form, and the animals die within 1 day of birth because of renal failure. Ureteric bud formation is impaired in glial cell line–derived neurotrophic factor (GDNF)

knockout mice, but p53 gene knockout mice do not demonstrate significant renal developmental anomaly.

6. **d. GDNF arrests the ureteric bud growth in vitro.** GDNF promotes ureteric bud growth in vitro. Although the importance of RET in kidney development was clearly demonstrated, it is only recently that its ligand, GDNF, has been identified. GDNF is a secreted glycoprotein that possesses a cystine-knot motif. GDNF is expressed within the metanephric mesenchyme prior to ureteric bud invasion, and ureteric bud formation is impaired in GDNF knockout mice.
7. **c. Urogenital sinus.** With use of the cell lineage studies the trigone was found to form mostly from bladder smooth muscle cells with only a minor contribution from the ureters. This is in contrast to the classic hypothesis where the trigone is formed from an extension of the ureteral development into the bladder.
8. **b. The median umbilical ligament.** By the 12th week, the urachus involutes to become a fibrous cord, which becomes the median umbilical ligament.
9. **d. Histologic evidence of smooth muscle differentiation begins near the bladder neck and proceeds toward the bladder dome.** Between the 7th and 12th weeks, the surrounding connective tissues condense and smooth muscle fibers begin to appear, first at the region of the bladder dome and later proceeding toward the bladder neck.
10. **b. Fifth week.** During the fifth week, primordial germ cells migrate from the yolk sac along the dorsal mesentery to populate the mesenchyme of the posterior body wall near the 10th thoracic level. In both sexes, the arrival of primordial germ cells in the area of future gonads serves as the signal for the existing cells of the mesonephros and the adjacent coelomic epithelium to proliferate and form a pair of genital ridges just medial to the developing mesonephros.
11. **b. In male embryos, the paramesonephric ducts degenerate under the influence of the MIS (müllerian-inhibiting substance) produced by the Leydig cells.** A new pair of ducts, called the paramesonephric (müllerian) ducts, begins to form just lateral to

the mesonephric ducts in both male and female embryos. These ducts arise by the craniocaudal invagination of thickened coelomic epithelium, extending all the way from the third thoracic segment to the posterior wall of the developing urogenital sinus. The caudal tips of the paramesonephric ducts adhere to each other as they connect with the urogenital sinus between the openings of the right and left mesonephric ducts. The cranial ends of the paramesonephric ducts form funnel-shaped openings into the coelomic cavity (the future peritoneum). As developing Sertoli cells begin their differentiation in response to the SRY (sex-determining region of the Y chromosome), they begin to secrete MIS, which causes the paramesonephric (müllerian) ducts to regress rapidly between the 8th and 10th weeks. Small müllerian duct remnants can be detected in the developed male as a small tissue protrusion at the superior pole of the testicle, called the appendix testis, and as a posterior expansion of the prostatic urethra, called the prostatic utricle. In female embryos, MIS is absent, so the müllerian ducts do not regress and instead give rise to fallopian tubes, uterus, and vagina.

12. **c. Prostate.** Vas deferens and appendix epididymis all develop from the mesonephric ducts. Testis from the gonadal ridge. The prostate and bulbourethral glands develop from the urogenital sinus.
13. **d. It requires the hormonal effects of MIS.** The prostate gland begins to develop during the 10th to 12th week as a cluster of endodermal evaginations budding from the pelvic urethra (derived from the urogenital sinus). These presumptive prostatic outgrowths are induced by the surrounding mesenchyme, and this process depends on the conversion of testosterone into dihydrotestosterone by 5α-reductase. Similar to renal and bladder development, prostatic development depends on mesenchymal-epithelial interactions but under the influence of androgens. There is no evidence that MIS plays a direct role in prostate development.
14. **c. Hymen.** In the absence of MIS and androgens, the mesonephric (wolffian) ducts degenerate and the paramesonephric (müllerian) ducts give rise to the fallopian tubes, uterus, and upper two-thirds of the vagina. The remnants of mesonephric ducts are found in the

mesentery of the ovary as the epoophoron and paroophoron, and near the vaginal introitus and anterolateral vaginal wall as Gartner duct cysts. The hymen develops from the endodermal membrane located at the junction between the vaginal plate and the definitive urogenital sinus, which is the future vestibule of the vagina.

15. **d. In females, the urethral folds become the labia majora, and the labioscrotal folds become the labia minora.** The early development of the external genital organ is similar in both sexes until 8 weeks gestation. Early in the fifth week, a pair of swellings called cloacal folds develops on either side of the cloacal membrane. These folds meet just anterior to the cloacal membrane to form a midline swelling called the genital tubercle. During the cloacal division into the anterior urogenital sinus and the posterior anorectal canal, the portion of the cloacal folds flanking the opening of the urogenital sinus becomes the urogenital folds, and the portion flanking the opening of the anorectal canal becomes the anal folds. A new pair of swellings, called the labioscrotal folds, then appears on either side of the urogenital folds. In the absence of dihydrotestosterone, the primitive perineum does not lengthen, and the labioscrotal and urethral folds do not fuse across the midline in the female embryo. The phallus bends inferiorly, becoming the clitoris, and the definitive urogenital sinus becomes the vestibule of the vagina. The urethral folds become the labia minora, and the labioscrotal folds become the labia majora. The external genital organ develops in a similar manner in genetic males who are deficient in 5α-reductase and therefore lack dihydrotestosterone.
16. **b. Third month.** The testicle reaches the level of internal inguinal ring by the third month and passes through the inguinal canal to reach the scrotum between the seventh and ninth months.
17. **d. It causes the regression of mesonephric ducts.** When the Y-linked master regulatory gene, called *SRY*, is expressed in the male, the epithelial cells of the primitive sex cords differentiate into Sertoli cells, and this critical morphogenetic event triggers subsequent testicular development. Analysis of DNA narrowed the location of the SRY to a relatively small region within the short arm of the chromosome. It is now clear that only about 25% of sex

reversals in humans can be attributed to disabling mutations of the SRY.

Chapter Review

1. The glomerulus, proximal tubule, loop of Henle, and distal tubule are derived from the metanephric mesenchyme.
2. The remainder of the collecting system is formed from the ureteric bud.
3. The Weigert-Meyer rule states that the most lateral and cephalad ureteric orifice arises from the lower pole and may demonstrate reflux whereas the most medial and caudad orifice drains the upper pole and may be associated with a ureterocele.
4. Sertoli cells produce müllerian-inhibiting substance, which causes regression of the müllerian ducts.
5. Testosterone is secreted by the Leydig cells and stimulates the wolffian ducts to form the vas deferens and seminal vesicles.
6. The prostate and bulbourethral glands develop from the urogenital sinus.
7. Circulating androgens play a critical role in the development of the prostate.
8. When 5α-reductase is deficient, prostatic growth and development is severely compromised.
9. In the absence of müllerian-inhibiting substance and androgens, the wolffian ducts degenerate and the müllerian ducts give rise to the fallopian tubes, uterus, and upper two-thirds of the vagina.
10. Boys with spina bifida have a 23% incidence of cryptorchidism.
11. If the *SRY* gene complex is translocated to an X chromosome, an XX female will have male characteristics.
12. The renin-angiotensin system is important for the normal development and growth of the kidney.
13. Circulating androgens and the conversion of testosterone to dihydrotestosterone (DHT) are critical to the normal development of the prostate and male external genitalia.

14. A defect in the WT1 gene may result in hypospadias, cryptorchidism, and ambiguous genitalia (disorders of sex development).
15. Defects in the androgen receptor result in abnormal masculinization of the external genitalia.
16. Abdominal pressure appears to be important for the transit of the testis through the inguinal canal and into the scrotum.
17. The embryonic kidneys are, in order of their appearance, the pronephros, the mesonephros, and the metanephros. The first two kidneys regress in utero, and the third becomes the permanent kidney.
18. Older, more differentiated nephrons are located in the inner part of the kidney near the juxtamedullary region.
19. Primordial germ cells migrate from the yolk sac along the dorsal mesentery to populate the mesenchyme of the posterior body wall near the 10th thoracic level.
20. Sertoli cells begin their differentiation in response to the SRY (sex-determining region of the Y chromosome); they begin to secrete MIS, which causes the paramesonephric (müllerian) ducts to regress.
21. Müllerian duct remnants in the male include the prostatic utricle and the appendix testis.

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Urologic Aspects of Pediatric Nephrology

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Keywords

acute kidney injury; chronic kidney disease; dialysis; end-stage renal disease (ESRD); glomerulonephritis; hematuria; hypertension; nephrotic syndrome; pediatric; proteinuria; renal transplantation

Questions

1. The mature nephron segments arise from which embryonic precursors?
    a. Pronephros
    b. Mesonephros
    c. Mesonephric duct
    d. Metanephros
    e. A combination of the mesonephros and metanephros
2. In term infants, which of the following are fully mature?
    a. Glomerular filtration rate (GFR)
    b. Tubular maturation
    c. Nephrogenesis
    d. Renal growth
    e. Renal sodium regulation
3. A 10-year-old healthy boy is found on routine physical examination to have asymptomatic microscopic hematuria. Urinalysis reveals

yellow urine with a specific gravity of 1.020, pH 6, no protein, and moderate amount of blood. The microscopic examination shows 5 to 6 red blood cells (RBCs) per high-powered field and no white blood cells, bacteria, or crystals. The most appropriate next step would be:

a. renal ultrasonography.
b. spot urine calcium and creatinine determinations.
c. urine culture.
d. measurement of serum creatinine.
e. repeat urinalysis in 2 weeks.

4. Urine protein excretion can be assessed qualitatively and quantitatively; which of the following represents abnormal urine protein excretion?

a. A term infant with a urine protein to creatinine ratio (UPC) of 0.6 mg/mg
b. A 13-year-old girl with a UPC of 1.0 mg/mg
c. A first morning urine sample with a UPC of 0.1 mg/mg
d. A 24-hour urine collection in a 15-year-old male with 150 mg of total protein
e. A 12-hour urine collection with 3.1 $mg/m^2/h$ of protein

5. A 6-year-old boy presents with generalized swelling and weight gain. His blood pressure is normal and his urinalysis notes clear yellow urine, a specific gravity of 1.030, pH 5.5, 3+ protein, and trace blood. Laboratory testing shows a creatinine of 0.2 mg/dL, an albumin of 1.8 gm/dL, a cholesterol of 350 mg/dL, and a spot UPC of 10 mg/mg. The most likely cause of his nephrotic syndrome is:

a. congenital nephrotic syndrome.
b. focal segmental glomerulosclerosis.
c. membranous nephropathy.
d. mesangial proliferation.
e. minimal change disease.

6. An 8-year-old boy is seen in the emergency room with new-onset headaches and gross hematuria. He has no history of fever, dysuria, joint pains, rash, or abdominal pain. He had a sore throat 2 weeks before, which resolved with antibiotics. His blood pressure is

125/90 mm Hg, and his physical examination is remarkable for 1+ lower extremity edema. The urinalysis notes tea-colored urine with a specific gravity = 1.025, pH = 6.5, large blood, and 2+ protein. His diagnostic evaluation should include all of the following EXCEPT:

a. computed tomography (CT) of the abdomen and pelvis.
b. comprehensive metabolic panel.
c. C3 determination.
d. antistreptolysin O (ASO)/DNase B titer.
e. microscopic examination of the urine.

7. The causes of hypocomplementemic nephritis in children include all of the following EXCEPT:

a. poststreptococcal glomerulonephritis.
b. membranoproliferative glomerulonephritis.
c. lupus nephritis.
d. antineutrophil cytoplasmic antibody (ANCA)–associated glomerulonephritis.
e. C3 glomerulopathy.

8. A 6-year-old girl is seen in the emergency room with complaints of abdominal and joint pains with gross hematuria. Her blood pressure is normal. Her physical examination is remarkable for a raised red rash on her lower legs and buttocks, mild abdominal pain on palpation, and bilateral ankle pain with minimal swelling. Her urinalysis noted tea-colored urine with a specific gravity of 1.020, pH 6.5, 3+ blood, and 2+ protein; urine microscopy showed numerous RBC cast. Her most likely diagnosis is:

a. Henoch-Schönlein purpura (HSP).
b. Postinfectious glomerulonephritis.
c. Hemolytic uremic syndrome (HUS).
d. Alport syndrome.
e. ANCA-associated glomerulonephritis.

9. A 3-year-old boy presents to clinic with failure to thrive. His blood pressure is normal. His initial evaluation is remarkable for a serum sodium of 144 meq/L, potassium of 2.5 meq/L, a chloride of 105 meq/L, a HCO_3 of 9 meq/L, and a phosphorus of 4.5 mg/dL. A

urinalysis notes a specific gravity = 1.005, pH = 7.0, and no blood or protein. A renal ultrasound shows normal renal size and shape with findings consistent with nephrocalcinosis. His most likely diagnosis is:

a. Fanconi syndrome.
b. proximal renal tubular acidosis (RTA).
c. Bartter syndrome.
d. Dent disease.
e. distal RTA.

10. A 4-year-old girl is referred to clinic because her mother was recently diagnosed with cystic renal disease and a polycystin-1 mutation. Likely findings on ultrasound in this child could include all of the following EXCEPT:

a. normal renal ultrasound.
b. a unilateral renal cyst.
c. hepatic fibrosis.
d. multiple unilateral cysts.
e. multiple bilateral renal cysts.

11. A 10-year-old boy is seen in the emergency department with the new onset of left-sided flank pain, gross hematuria, and vomiting. His blood pressure is 100/65 mm Hg, and the physical examination reveals right costovertebral angle tenderness. The urinalysis shows brown urine with a specific gravity = 1.030, pH = 7, large amount of blood, and 2+ protein. The next step in diagnostic evaluation of choice should be:

a. high-resolution CT of abdomen/pelvis without contrast.
b. microscopic examination of the urine.
c. cystoscopy.
d. abdominal ultrasound.
e. serum electrolyte determination.

12. Common causes of hyperoxaluria include all of the following EXCEPT:

a. vitamin C excess.
b. short gut syndrome.
c. oxylate-rich diets.
d. alanine:glyoxylate aminotransferase deficiency.

e. adenine phosphoribosyltransferase (APRT) deficiency.

13. A 15-year-old boy has a blood pressure of 120/80 mm Hg during routine examination. The child is calm, and a properly sized blood pressure cuff was used for blood pressure measurement. The family history is strongly positive for essential hypertension and diabetes mellitus. His physical examination is unremarkable except for a body mass index (BMI) of 29 kg/m^2. The best next step in management is to:
    a. repeat blood pressure measurement in 1 to 2 weeks.
    b. obtain a fasting lipid profile.
    c. perform Doppler renal ultrasonography.
    d. obtain peripheral vein renin levels.
    e. obtain 24-hour ambulatory blood pressure monitoring (ABPM).
14. A 12-year-old boy is sent to the emergency department for evaluation of vomiting and decreased urination. He has a blood urea nitrogen (BUN) level = 40 mg/dL and a creatinine = 1.8 mg/dL. His urinary sodium level = 20 meq/L, fractional excretion of sodium (FE_{Na}) = 0.8%, and urinary osmolality = 600 mOsm/kg. The most likely cause of his elevated creatinine is:
    a. posterior urethral valves.
    b. dehydration.
    c. interstitial nephritis.
    d. Hemolytic uremic syndrome.
    e. acute urinary tract infection (UTI).
15. Common complications of advanced chronic kidney disease in children include all of the following EXCEPT:
    a. growth hormone deficiency.
    b. protein malnutrition.
    c. erythropoietin deficiency.
    d. metabolic acidosis.
    e. 1,25 vitamin D insufficiency.
16. A 5-year-old-boy with posterior urethral valves since birth has renal function that has slowly declined, with a current creatinine = 1.7

mg/dL. His growth is impaired, with height and weight less than the third percentile. The most appropriate management strategy is:
   a. referral for initiation of transplant evaluation.
   b. urologic evaluation for urethral obstruction.
   c. repeat urodynamics in expectation of preemptive transplantation.
   d. optimization of nutritional intake and recombinant growth hormone therapy.
   e. G-tube placement.

17. The most common cause of pediatric renal allograft loss is:
   a. primary anastomotic failure.
   b. recurrent obstructive uropathy.
   c. antibody mediated rejection.
   d. acute cellular rejection.
   e. chronic allograft nephropathy.

Answers

1. **d. Metanephros.** The metanephros forms the mature nephron segments. The pronephros undergoes complete involution, while residual segments of the mesonephros, the mesonephric duct, forms portions of the urinary tract.
2. **c. Nephrogenesis.** Nephrogenesis is complete by 34 to 36 weeks post conception, whether in utero or ex utero. Tubular maturation, including urinary sodium regulation, continues for weeks to months post partum. GFR reaches adult levels by 2 years of age, whereas renal growth continues until the postpubertal period.
3. **e. Repeat urinalysis in 2 weeks.** In a healthy asymptomatic child, isolated microscopic hematuria can occur in up to 1% of school-age children. With repeat measure, less than 25% will have persistent hematuria. Should the microscopic hematuria be persistent, then screening studies of renal function, renal structure, urinary calcium excretion, and possible infection are indicated.
4. **b. A 13-year-old girl with a UPC of 1.0 mg/mg.** A UPC of greater than 1.0 mg/mg is abnormal. Normal urine protein excretion as

measured by UPC is less than 0.7 mg/m in a newborn and less than 0.2 mg/mg in older children. A 24-hour protein collections in adolescents and adults should be less than 200 mg/24 h or in younger children less than 4 mg/m^2/h. A UPC of greater than 2.0 mg/mg defines nephrotic range proteinuria.

5. **e. Minimal change disease.** The most common cause of nephrotic syndrome in childhood is minimal change disease. Congenital nephrotic syndrome by definition is seen in children younger than 3 months of age. Membranous nephropathy and mesangial proliferation are seen less frequently in childhood, whereas focal segmental glomerulosclerosis (FSGS) become increasing common during adolescence and late childhood.
6. **a. Computed tomography (CT) of the abdomen and pelvis.** A child presenting with symptomatic hypertension and gross hematuria is strongly suggestive of acute glomerulonephritis. The prior history of pharyngitis is consistent with poststreptococcal-associated disease. Examination of the urine sediment for signs of glomerulonephritis (cellular casts) and documentation of renal function and electrolytes, as well as elevated antistreptolysin O titer and decreased C3, are recommend to confirm the diagnosis. Abdominal CT scans play no role in the initial evaluation of suspected acute glomerulonephritis.
7. **d. Antineutrophil cytoplasmic antibody (ANCA)–associated glomerulonephritis.** Serum complement levels are very useful in narrowing the differential diagnosis in acute glomerulonephritis. Low C3 and C4 levels are seen in lupus nephritis and membranoproliferative glomerulonephritis (MPGN), whereas low C3 and normal C4 levels are seen in postinfectious glomerulonephritis (PIGN) and C3 glomerulonephritis (C3GN). Complement levels are normal in patients with ANCA-associated glomerulonephritis.
8. **a. Henoch-Schönlein purpura (HSP).** HSP is a systemic vasculitis that is characterized by purpuric rash, commonly on legs and buttocks, abdominal pain, arthritis, and nephritis. HUS is commonly seen in association with bloody diarrhea. Alport

syndrome is not commonly seen with systemic complaints, whereas PIGN is usually preceded by either pharyngitis or impetigo. ANCA-associated glomerulonephritis, although rare is childhood, can be seen in older children or adolescents, and systemic complaints including fever are common.

9. **e. Distal RTA.** Failure to thrive in an infant in association with hyperchloremic acidosis (nonanion gap) is strongly suggestive of RTA. Nephrocalcinosis is a common finding in distal RTA, as is the very low HCO_3 level. With proximal RTA or Fanconi, HCO_3 levels are typically in the 12 to 14 meq/L range, and hypophosphatemia, low-grade proteinuria, and glycosuria are commonly seen. Dent disease does not present with RTA, and Bartter syndrome is defined by metabolic alkalosis.
10. **c. Hepatic fibrosis.** Hepatic fibrosis is a feature commonly seen in autosomal recessive polycystic kidney disease (ARPKD). Mutations in polycystin-1 are associated with autosomal dominant polycystic kidney disease (ADPKD). Renal ultrasound findings in childhood can be highly variable and range from “normal” to severe bilateral cystic disease.
11. **d. Abdominal ultrasound.** This clinical scenario is most consistent with acute presentation of a renal or ureteral stone. Ultrasound imaging to rule out hydronephrosis and to identify presence of a stone is the best first step. Intervention for drainage is not indicated until appropriate analgesics have been initiated. CT imaging is unlikely to add further information and exposes the child to radiation.
12. **e. Adenine phosphoribosyltransferase (APRT) deficiency.** Hyperoxaluria can be primary, in the case of alanine:glyoxylate aminotransferase deficiency, or secondary due to excessive oxalate intake, vitamin C (a precursor of oxalate), or hyperabsorption in the case of short gut syndrome. APRT deficiency is an autosomal recessive disorder of adenine metabolism linked with recurrent dihydroxyadenine stones.
13. **a. Repeat blood pressure measure in 1 to 2 weeks.** Repeat the measurement next week in the office. This child’s blood pressure is

elevated, and measurement should be repeated and possible hypertension confirmed. ABPM could be considered and is recommended; however, availability is a challenge and is more practically performed after repeated office measurements. Therefore the blood pressure should be remeasured in the office. If hypertension is confirmed, the workup will include a complete metabolic panel, lipid profile, echocardiogram, and renal ultrasonography. Angiography and selective renin levels can be considered in case of severe or recalcitrant hypertension.

14. **b. Dehydration.** The initial evaluation of patients with suspected acute kidney injury (AKI) should include urinary indices including urine sodium, osmolality, and creatinine. A low urine sodium (<25 meq/L) or a FeNa of less than 1% is consistent with "prerenal" AKI, or renal hypoperfusion as seen with dehydration of vomiting and/or diarrhea. An elevated BUN and creatinine due to posterior urethral valves, interstitial nephritis, or UTIs are due to "intrinsic" renal disease, and urinary sodium and FeNa (>1%) are typically elevated.
15. **a. Growth hormone deficiency.** Patients with chronic kidney disease (CKD) have growth failure due to abnormalities in growth hormone activity rather than true growth hormone deficiency. Protein malnutrition, anemia from erythropoietin deficiency, chronic metabolic acidosis, and an inability to convert 25 vitamin D to active 1,25 vitamin D are all common feature of advanced CKD in children.
16. **d. Optimization of nutritional intake and recombinant growth hormone therapy.** Impairment of growth is a common feature of CKD even at an early stage, regardless of etiology. Aggressive management of nutritional needs is often insufficient, and therapy with recombinant growth hormone is indicated in the majority of patients.
17. **e. Chronic allograft nephropathy.** The most common cause of renal transplantation loss in children is chronic allograft nephropathy. Primary anastomotic failure is very rare, and recurrent obstructive uropathy in most patients can be managed with medical or surgical interventions. Acute cellular rejection,

although common, is typically responsive to immunomodulation. Antibody-mediated rejection is increasing being recognized as a cause of graft failure.

Chapter Review

1. Congenital anomalies of the kidney and urinary tract (CAKUT) are the most common causes of end-stage renal disease (ESRD) in children.
2. Prenatal and neonatal GFR is significantly lower than adult GFR levels.
3. The maintenance of the balance of sodium is an important function of the neonatal kidney.
4. The newborn kidney has a limited capacity to excrete potassium and tends to have higher serum potassium values than older children.
5. Isolated microscopic hematuria is reported to occur in approximately 1% of school-age children.
6. Most children with microscopic hematuria will have spontaneous resolution within 1 year of presentation.
7. Gross hematuria and a urinalysis with no RBCs but a positive heme dipstick is strongly suggestive of hemoglobinuria or myoglobinuria.
8. It is important to recognize that urine contains protein under physiologic conditions and especially with fever and exercise.
9. Transient proteinuria accounts for more than 75% of patients with isolated proteinuria.
10. If a patient with proteinuria has evidence of systemic disease or glomerular nephritis, testing should include C3, C4, and antinuclear antibody (ANA) screening.
11. Corticosteroids will induce a remission in the vast majority of children with nephrotic syndrome.
12. Focal segmental glomerulosclerosis is seen in the majority of children with steroid-resistant nephrotic syndrome.
13. Minimal change nephrotic syndrome is the most common cause of nephrotic syndrome in childhood.

14. Immunoglobulin A (IgA) nephropathy is the most common primary cause of glomerulonephritis throughout the world.
15. HUS is a common cause of severe AKI in a previously healthy young child.
16. The hallmark of **RTA** is hyperchloremic metabolic acidosis (normal anion gap acidosis).
17. Children with distal RTA classically have hyperchloremic metabolic acidosis, hypokalemia, hypercalciuria, and nephrocalcinosis.
18. Hypokalemic metabolic alkalosis is the defining feature of Bartter syndrome.
19. RTA associated with hyperkalemia is also known as type 4 RTA.
20. Urologic anomalies are seen in up to 25% of patients with multicystic dysplastic kidney (MCDK), including renal hypoplasia/dysplasia, vesicoureteral reflux (VUR), and ureteropelvic junction (UPJ) obstruction.
21. ARPKD is caused by a mutation in the PKHD1 gene.
22. ADPKD is due to mutations in the polycystin-1 (PKD-1) or polycystin-2 (PKD2) genes.
23. Urolithiasis remains underdiagnosed because many children are asymptomatic at the time of diagnosis.
24. Calcium stones are currently the most prevalent stones in children.
25. Cystinuria results in decreased tubular reabsorption of cystine, ornithine, arginine, and lysine.
26. The increase in the prevalence in pediatric hypertension in recent years is closely linked to increasing obesity rates in children.
27. Up to a quarter of patients have normal blood pressures outside of the office setting (“white coat” hypertension).
28. Therapeutic lifestyle changes are the initial intervention of choice for patients with primary hypertension without end-organ damage.
29. Hospitalized pediatric patients with AKI have a nearly 15-fold increased mortality risk compared with patients without AKI.
30. Prerenal AKI requires prompt and vigorous fluid resuscitation with isotonic fluids.
31. Continuous renal replacement therapy (CRRT) has become the standard of care because it delivers slow continuous correction of fluid overload and electrolyte imbalances.

32. Causes of CKD in children are very different from those that cause CKD in adults and vary by age group within the pediatric population.
33. The most common cause of anemia with CKD is a deficiency of erythropoietin.
34. With CKD, kidney function declines in a nonlinear fashion, as the later stages are associated with a more rapid progression.
35. Peritoneal dialysis is the most common initial modality for dialysis in children younger than 9 years of age and less than 20 kg.
36. The unique benefits of renal transplantation include improved growth, neurocognitive development, and quality of life.
37. The risk for acute cellular rejection in the first year after pediatric renal transplantation is less than 10% in living donor transplants.
38. The risk for acute cellular rejection in the first year post transplant is approximately 15% in deceased donor transplants.

22

Perinatal Urology

C.D. Anthony Herndon, and Rebecca S. Zee

Keywords

prenatal urinary tract dilation; UPJ obstruction; urinary tract infection; VUR

Questions

1. A newborn girl has prenatally detected left A1 urinary tract dilation (UTD) with 8-mm anteroposterior diameter APD on third-trimester US. A renal ultrasound (US) is obtained on day of life (DOL) 3, which demonstrates left P1 UTD. What is the most likely cause of this condition?
    a. Vesicoureteral reflux (VUR)
    b. Transient dilation
    c. Ureteropelvic junction (UPJ) obstruction
    d. Ureterovesical junction (UVJ) obstruction
    e. Ectopic ureteral insertion
2. A circumcised newborn boy with prenatally detected A2-3 UTD receives a renal ultrasound on DOL 3 that is normal. What is the best next step?
    a. Discharge from urologic care
    b. Voiding cystourethrogram (VCUG)
    c. Repeat US in 3 to 6 months
    d. Start prophylactic antibiotics

e. Renal scintigraphy

3. Which of the following is accurate with regard to prenatal intervention for lower urinary tract obstruction (LUTO)?
    a. Improved survival; unchanged long-term renal function
    b. Decreased survival; improved long-term renal function
    c. Unchanged survival; improved long-term renal function
    d. Improved survival; decreased long-term renal function
    e. Decreased survival; unchanged long-term renal function
4. A fetus is diagnosed with lumbosacral myelomeningocele. After having an informed discussion of risks and benefits, the pregnant mother would like you to discuss the urologic benefits of prenatal intervention with her. Which of the following is the best answer?
    a. Prenatal intervention improves urodynamic parameters, need for clean intermittent catheterization (CIC), and reconstruction in most patients who undergo prenatal myelomeningocele closure.
    b. Prenatal intervention has not shown a benefit to urodynamic parameters, nor does it decrease the need for CIC and for urinary reconstruction.
    c. The risks outweigh the benefits, and prenatal intervention should never be performed.
    d. Prenatal intervention for myelomeningocele can be done at any center that has a pediatric neurosurgeon and pediatric urologist.
    e. Postnatal intervention is associated with a lower incidence of preterm labor.
5. A 6-week-old circumcised boy is referred to your office for A1 prenatal UTD. His postnatal US at DOL 2 demonstrated central calyceal dilation with an APD of 11 mm (P1). The ureter and bladder appeared normal. An US repeated today is unchanged (P1). He is otherwise healthy. What is his most likely outcome?
    a. He has a UPJ obstruction and requires surgery.
    b. He should have a VCUG and a technetium-mercaptoacetyltriglycine (Tc-MAG3) Lasix renal scan before further discussion.
    c. His UTD is most likely to resolve within the next 2.5 years.

d. His UTD will resolve, but the timing is unknown.
e. He should start prophylactic antibiotics to prevent an infection.

6. Which of the following is correct with regard to VUR and UTD?
    a. All patients who present with P2 UTD should undergo VCUG.
    b. UTD correlates in a linear fashion with VUR.
    c. All patients with VUR will demonstrate UTD to some degree on postnatal US.
    d. VUR will occur more commonly in females.
    e. VCUG is recommended in patients with increased risk of UTI.
7. Which of the following is TRUE regarding the link between UTD and UTI?
    a. There is no correlation between degree of dilation and risk of UTI.
    b. Gender does not impact risk of UTI.
    c. Recent data fail to demonstrate the efficacy of prophylactic antibiotics to prevent UTI in patients with UTD.
    d. All patients with UTD should be started on prophylactic antibiotics at birth.
    e. The event rate of UTI in patients with UTD is high.
8. A newborn with P3 UTD undergoes a MAG3 Lasix renal scan at 3 months of age. Which of the following is TRUE?
    a. Indications for surgery include decreased differential function (<40%), delayed t½ (>20 minutes), and significant retention on delayed upright imaging.
    b. MAG3 Lasix renal scan provides sufficient anatomic detail to identify the insertion point of an ectopic ureter.
    c. A MAG3 Lasix renal scan is an accurate measure of glomerular filtration rate (GFR).
    d. Dehydration does not impact the results of the study.
    e. Furosemide (Lasix) should be given at the exact same time interval after radionuclide administration for every patient.
9. An uncircumcised newborn boy with UTD A2-A3 presents to your clinic at 4 weeks of age. He is on amoxicillin and has not experienced any UTIs. He has been otherwise healthy. A repeat US today demonstrates peripheral calyceal dilation, APD 20 mm, 10-

mm dilation of distal right ureter, and a normal bladder. VCUG is negative for reflux. What is the next best step?

a. Transition to trimethoprim/sulfamethoxazole prophylaxis
b. Perform MAG3 scan
c. Repeat US in 3 months
d. Recommend newborn circumcision
e. Stop prophylaxis

10. Pediatric urology is urgently consulted for evaluation of an abnormal scrotum in a 2-day-old neonate. The primary team reports that the newborn examination was normal at birth and has changed in the past 2 hours. On exam, the right testis is tender, firm, and overlying skin is discolored. The left testis is normal. The neonate is fussy. What is the next step?

a. Elective contralateral septopexy
b. Scrotal US with Doppler
c. Initiate antibiotics
d. Serial scrotal examination
e. Emergent surgical exploration

Answers

1. **b. Transient dilation.** A majority of UTD is transient (41%–88%). Normal values for UTD are less than 7 mm at greater than 28 weeks. This prenatal US with 8-mm anteroposterior diameter of renal pelvic diameter (APRPD) in the third trimester would represent low risk, or A1 disease. UPJ obstruction is the second most common cause of UTD. VUR accounts for 10% to 20% UTD and other causes account for less than 10% of UTD. A postnatal US after 48 hours of age is recommended.
2. **c. Repeat US in 3 to 6 months.** If the first postnatal US demonstrates no dilation after 48 hours of age, a US should be repeated in 3 to 6 months to confirm that UTD has resolved. Up to 28% of UTD may be detected after initial negative US. An informed discussion should be performed with the family to discuss therapeutic options. VCUG should not be performed as the risk of

UTI from reflux is currently low in the absence of dilation in a circumcised boy. Similarly, prophylactic antibiotics and renal scintigraphy are not warranted given the low risk of infection.

3. **a. Improved survival; unchanged long-term renal function.** Several publications have shown that prenatal intervention for LUTO improves survival without a significant effect on renal function. Prenatal intervention for LUTO is most beneficial for those patients who present with oligohydramnios early in the second trimester to facilitate pulmonary development after 27 weeks' gestation.
4. **b. Prenatal intervention has not shown a benefit to urodynamic parameters, nor does it decrease the need for CIC and for urinary reconstruction.** A multicenter trial was published in 2011 that randomized patients to prenatal closure versus postnatal closure of myelomeningocele. The study reported that prenatal closure decreased the need for ventriculoperitoneal shunting, as well as motor and cognitive functioning. However, prenatal closure did not affect urodynamic parameters or the need for CIC or urinary reconstruction compared to postnatal closure.
5. **c. His UTD is most likely to resolve within the next 2.5 years.** Most patients with P1 UTD will resolve within 30 months of birth. This patient has P1 UTD and should be observed with serial ultrasound alone. Guidelines suggest that VCUG should be performed at the discretion of the clinician but should not universally be applied. UPJ obstruction is unlikely with low-risk upper tract dilation. Prophylactic antibiotics and additional imaging studies such as VCUG and MAG3 scans should be reserved for those patients who are at the highest risk of infection, which includes females, uncircumcised males, high-grade dilation (SFU [Society of Fetal Urology] grades 3–4), and ureteral dilation.
6. **e. VCUG is recommended in patients with increased risk of UTI.** VCUG is recommended for any patient with P3 UTD or high risk. P2 and P3 risk categories both include an APRPD greater than or equal to 15 mm, peripheral calyceal dilation, or ureteral abnormality. P2 patients may undergo VCUG at discretion of the clinician. P3 or high risk is distinguished by any abnormality of

renal parenchymal thickness, renal parenchymal appearance, or abnormality of the bladder. Therefore a patient with P3 UTD or high risk should undergo a follow-up US at 1 month and a VCUG and should be started on prophylactic antibiotics. Renal scintigraphy can be performed at the clinician's discretion. VCUG can be performed on an outpatient basis. VUR will be present in up to 31% of patients screened with UTD, will occur more commonly in males, and will be high grade and bilateral. Upper tract dilation correlates poorly with presence of VUR. One-quarter of patients with VUR on postnatal VCUG will have no dilation on a postnatal US.

7. **c. Recent data fail to demonstrate the efficacy of prophylactic antibiotics to prevent UTI in patients with UTD.** A recent meta-analysis of 11 studies failed to show a benefit of continuous prophylactic antibiotics (CAP) on preventing UTI in patients with prenatal UTD. The degree of UTD is predictive of risk of UTI. Risk factors for UTI include female gender, uncircumcised status, and higher grades of UTD. PA are recommended only for those at higher risk (P2 and P3). The incidence of UTI is low (8%–19%), which makes it difficult to study UTI rates in the UTD population.
8. **a. Indications for surgery include decreased differential function (<40%), delayed t½ (>20 minutes), and significant retention on delayed upright imaging.** Sussman et al. reported that indications for surgery in an otherwise asymptomatic patient include decreased split differential function, delayed drainage, or significant retention on delayed upright imaging. A limitation of a MAG3 renal scan is that it does not provide high-resolution imaging of the genitourinary (GU) system. The best imaging modality to assess the insertion point of an ectopic ureter would be magnetic resonance (MR) urogram. MAG3 is not filtered through the glomerulus. Rather, it is handled by the proximal tubule and therefore is not an accurate measure of GFR. Dehydration of the patient may affect the results of a MAG3 scan. Administration of Lasix should be individualized for each patient at the height of tracer uptake within the collecting system.

9. **d. Recommend newborn circumcision.** Uncircumcised status and ureteral dilation are risk factors for UTI in this male patient with P2 intermediate risk UTD. An informed discussion should be held with the family to discuss risk of UTI, risk of circumcision, and continued prophylaxis. The patient is too young to be transitioned to Bactrim, which should not be administered prior to 2 to 3 months of age. Similarly, prophylactic antibiotics should not be discontinued in this patient. Renal scintigraphy may be performed at the clinician's discretion, but recommendation for circumcision should not be delayed to assess renal drainage and function.
10. **e. Emergent surgical exploration.** An acute postnatal change in scrotal examination suggests a postnatal torsion. Up to 44% of postnatal torsion can be salvaged with emergent exploration and septopexy. In prenatal torsion, the postnatal exam will not change from birth. The testis will be a solid, nontender mass and demonstrate absence of flow on Doppler US. The affected testis is not salvageable in prenatal torsion. Most pediatric urologists would perform contralateral septopexy on an urgent basis in patients with a diagnosis of prenatal torsion to prevent asynchronous contralateral torsion. Scrotal US with Doppler may have a role if the diagnosis is unclear, but surgical exploration should not be delayed if the clinician has made a diagnosis of postnatal torsion. Serial scrotal examinations are not warranted, because postnatal torsion can be diagnosed clinically and should be addressed as a surgical emergency. Likewise, there is no role for antibiotics, because this clinical picture is not consistent with an infectious process.

Chapter Review

1. The initial screening maternal ultrasound is performed at approximately 20 weeks' gestation. Low-risk pregnancies average two scans and high-risk four scans throughout pregnancy.
2. Urinary tract dilation represents the second most common condition detected prenatally and occurs in 1% to 3% of pregnancies.

3. Ultrasound cannot reliably predict the severity of disease for all urologic conditions. It tends to correlate best with obstructive lesions such as UPJ obstruction or posterior urethral valves (PUV) and less with vesicoureteral reflux.
4. Prenatal detection of urologic disease affords the opportunity for the family to meet with a specialist prior to delivery to become informed of the anticipated course and natural history of the suspected diagnosis.
5. An inverse relationship exists between the severity of UTD and incidence of urologic disease.
6. APD demonstrates a linear relationship for obstructive lesions such as UPJ obstruction and PUV.
7. An APD of 15 in the third trimester demonstrates reasonable positive predictive value for the need for surgery for UPJ obstruction.
8. The SFU hydronephrosis grading system consists of a five-point grading system that is based on calyceal dilation and integrity of the parenchyma.
9. The UTD grading system consists of a combination of APD and SFU systems with the additional inclusion of the lower urinary tract and uses a standardized reporting system and risk stratification.
10. Prenatal intervention for LUTO appears to improve perinatal survival but does not reliably impact long-term renal function.
11. Parental consultation for fetal myelomeningocele closure should take place at a center with expertise in fetal surgery that offers a multidiscipline approach.
12. Fetal surgery for myelomeningocele appears to improve motor/cognitive skills and decrease the need for VP shunting but does not offer definitive benefit to bladder function.
13. Clinical predictors of poor outcome are oligohydramnios less than 27 weeks EGA, renal cortical cysts, and increased echogenicity.
14. Prenatal intervention for LUTO appears to improve perinatal survival but does not reliably impact long-term renal function.
15. Fetal cystoscopy affords the opportunity to establish the diagnosis and provide treatment for LUTO.
16. Fetal cystoscopy may offer a short-term advantage for overall renal function when compared with vesicoamniotic shunting, but long-

term renal data do not exist.

17. The initial postnatal US should be obtained at 48 hours to account for intravascular depletion that occurs in the initial postnatal period secondary to third spacing, which later becomes mobilized to the intravascular space.
18. UTD classification assigns three levels of risk: P1 UTD (SFU 2) (low risk) APD 10 < 15 mm, P2 UTD (SFU 3) (intermediate risk) APD greater than 15 mm and peripheral calyceal dilation, P3 (SFU 4) (high risk) APD greater than 15 mm and/or positive values for other data points.
19. The degree of APD dilation correlates poorly with the presence and degree of VUR in patients with prenatal UTD.
20. Recommendations for VCUG for prenatal UTD should be based on risk of UTI in the absence of concerns for LUTO.
21. Complete resolution of P1 UTD should occur in most patients by 4 years of age.
22. For P1 UTD (SFU 2), PA and VCUG are not recommended.
23. For P2 UTD (SFU 3), the use of PA and VCUG is left to the discretion of the physician.
24. For P3 UTD (SFU 4), PA and VCUG are recommended, as well as a MAG3 renal scan.
25. The degree of dilation is predictive of risk of UTI in patients with prenatal UTD.
26. Risk factors for UTI for prenatal UTD include female, intact foreskin, high-grade kidney dilation, ureteral dilation, and VUR.
27. Prophylactic antibiotics are recommended for those at risk of UTI, including P2 UTD (SFU 3) and P3 UT (SFU 4).
28. Most patients with prenatal UTD will have transient dilation with reported resolution rates between 41% and 88%.
29. Prenatal APD dilation of 15 mm in third semester is predictive of the need for surgery.
30. VUR is reported in up to 31% of patients with prenatal UTD and will occur most commonly in males that have bilateral high-grade disease.
31. Multicystic dysplastic kidney (MCDK) does not pose an increased risk to Wilms tumor and can be safely observed in most patients.

32. Duplication anomalies may be associated with VUR to the lower pole moiety, which may influence decisions for intervention to the obstructed upper pole moiety.
33. Megaureter will resolve spontaneously in a majority of patients. However, surgical intervention is indicated for recurrent UTI, decreased function, or increasing UTD. Temporizing diverting procedures are recommended for younger than 1 year of age.
34. Fetal closure of myelomeningocele appears to increase motor/cognitive skills and decrease the need for a VP shunt, but it does not improve urologic symptoms and urodynamic parameters or decrease the need for urinary reconstruction.
35. A urologic source for abdominal distention in the newborn includes urinary ascites, urinary obstruction (upper and lower tract), renal mass (cystic and solid), and, rarely, a retroperitoneal mass (adrenal).
36. Urinary retention may occur with LUTO and may occur in males with PUV or females with urogenital (UG) sinus or cloacal anomaly.
37. Renal tumors are uncommon in the newborn period, and the most common is congenital mesoblastic nephroma.
38. The newborn with ambiguous genitalia (including any newborn "male" with bilateral nonpalpable testes) should prompt an evaluation for electrolyte disturbance due to classical salt wasting congenital adrenal hyperplasia.
39. Gross hematuria in the neonate is not common and should be evaluated with a urinalysis and Doppler abdominal ultrasound.
40. The acuity of neonatal testicular torsion will dictate the need for emergent exploration.

SECTION B

Basic Principles

OUTLINE

23. Urologic Evaluation of the Child

24. Pediatric Urogenital Imaging

25. Infection and Inflammation of the Pediatric Genitourinary Tract

26. Core Principles of Perioperative Management in Children

27. Principles of Laparoscopic and Robotic Surgery in Children

23

Urologic Evaluation of the Child

Rachel Selekman, and Hillary L. Copp

Keywords

diagnostic imaging; medical history taking; pediatrics; physical examination; urology

Questions

1. Which one of the following patients does NOT need to be seen emergently?
    a. A newborn with hydronephrosis in a solitary kidney
    b. A 4-year-old boy with acute right scrotal pain
    c. A 12-year-old girl with microscopic hematuria found during a routine examination
    d. An 8-year-old boy with sickle cell anemia and a 5-hour history of priapism
    e. A male newborn with a distended bladder, bilateral hydronephrosis, and respiratory insufficiency
2. Which of the following is a potential complication of neonatal circumcision?
    a. Wound infection
    b. Meatal stenosis
    c. Cicatrix
    d. Death
    e. All of the above

3. The pediatric kidney is particularly susceptible to trauma due to:
    a. relatively increased renal size.
    b. limited visceral adipose tissue.
    c. limited chest wall protection.
    d. increased mobility.
    e. all of the above.
4. What is the optimal timing of spinal ultrasonography during screening for occult spinal dysraphism?
    a. Before 6 months of age
    b. 6 months to 2 years of age
    c. At any age before puberty
    d. At any age
    e. Never. Ultrasound (US) is not useful to screen for spinal dysraphism.
5. What is the most commonly detected etiology for asymptomatic microscopic hematuria in children?
    a. Fibroepithelial polyp
    b. Hypercalciuria
    c. Poststreptococcal glomerulonephritis
    d. Uncomplicated urinary tract infection
    e. Hyperuricosuria
6. Findings associated with the Beckwith-Wiedemann syndrome include:
    a. macroglossia.
    b. hepatosplenomegaly.
    c. nephromegaly.
    d. macrosomia.
    e. all of the above.
7. A voiding cystourethrogram (VCUG) is essential in the diagnosis of which clinical condition?
    a. Ureteropelvic junction obstruction
    b. Primary obstructive megaureter
    c. Posterior urethral valves
    d. Nephrolithiasis
    e. Renal mass

8. When should a child with suspected congenital adrenal hyperplasia be tested?
    a. Before discharge from the nursery
    b. At the first well-baby visit
    c. At 6 months of age
    d. At puberty
    e. Only if undergoing general anesthesia
9. All of the following statements about the pediatric abdominal examination are true EXCEPT:
    a. renal pathology is the source of as many as two-thirds of neonatal abdominal masses.
    b. abdominal distention at birth or shortly afterward suggests either obstruction or perforation of the gastrointestinal tract.
    c. the abdominal wall is normally strong, especially in infants with hydronephrosis.
    d. a solid flank mass may be due to renal venous thrombosis.
    e. in cloacal exstrophy, an omphalocele is superior to the cecal plate and lateral bladder halves with prolapsed ileum in the midline.
10. Which of the following statements is FALSE about cutaneous markers of occult spinal dysraphism?
    a. Forty percent of patients with atypical presacral dimples have associated occult spinal dysraphism.
    b. A combination of two or more congenital midline skin lesions is the strongest marker of occult spinal dysraphism.
    c. A presacral dimple less than 2.5 cm from the anal verge at birth may indicate spina bifida or cord tethering.
    d. Sacral hypertrichosis may be associated with spinal dysraphism.
    e. All of the above are true.
11. Sexual abuse can be associated with which of the following physical examination findings?
    a. Bruised vaginal mucosa in a prepubertal child
    b. Penile discharge
    c. A normal genital and perineal examination
    d. a and c

e. a, b, and c

12. Urethral meatal stenosis in the infant occurs most commonly:
    a. as a result of birth trauma.
    b. after urinary tract infection.
    c. after a VCUG.
    d. after healing of the inflamed, denuded glans after circumcision.
    e. from penile adhesions.
13. In newborns with ambiguous genitalia, palpation of a gonad rules out which difference of sexual development (DSD/intersex)?
    a. Ovotesticular disorder
    b. Mixed gonadal dysgenesis
    c. Partial androgen insensitivity
    d. Pure gonadal dysgenesis
    e. Persistent müllerian duct syndrome
14. Secondary urinary incontinence is defined as:
    a. diurnal and nocturnal enuresis.
    b. incontinence associated with urinary tract infection.
    c. urinary incontinence associated with constipation.
    d. urinary incontinence after a dry interval greater than 6 months.
    e. urinary incontinence associated with a neurologic condition.
15. A newborn should have a scrotal hydrocele surgically corrected in the neonatal period if:
    a. it is large.
    b. it is changing in volume.
    c. it accompanies a symptomatic hernia.
    d. a, b, and c.
    e. b and c.

Answers

1. **c. A 12-year-old girl with microscopic hematuria found during a routine examination.** In the absence of other symptoms, microscopic hematuria in children is not an emergency. Bilateral hydronephrosis or hydronephrosis in a solitary kidney represents

neonatal emergencies and should be evaluated as soon as possible. Acute scrotal pain should always be considered testicular torsion until proven otherwise. Boys with sickle cell anemia are at increased risk for priapism and should always be treated immediately to decrease the long-term sequelae associated with priapism.

2. **e. All of the above.** Wound infections, meatal stenosis, removal of too much/too little prepuce, cicatrix, and even death are all potential complications of neonatal circumcision.
3. **e. All of the above.** The pediatric kidney is particularly susceptible to trauma due to limited visceral adipose tissue, limited chest wall protection, relatively increased renal size, and increased mobility of the kidney.
4. **a. Before 6 months of age.** Ossification of the posterior elements after 6 months of age prevents an acoustic US window. After 6 months, spinal magnetic resonance imaging (MRI) is recommended when an occult spinal dysraphism is suspected.
5. **b. Hypercalciuria.** Most microscopic hematuria in children is transient and the source is not identified. The most commonly identified etiology of asymptomatic microhematuria in children is hypercalciuria.
6. **e. All of the above.** Beckwith-Wiedemann syndrome is caused by a mutation on chromosome 11p15.5. Clinical features include macroglossia, nephromegaly, organomegaly (hepatosplenomegaly), macrosomia (gigantism), and hemihypertrophy. Many of the affected infants have hypoglycemia in the first few days of life. Patients are at increased risk for specific tumors (e.g., adrenal carcinoma, Wilms tumor, hepatoblastoma, and rhabdomyosarcoma).
7. **c. Posterior urethral valves.** The diagnosis of posterior urethral valves requires visualization of the urethra during voiding. Bladder diverticula, a pronounced bladder neck, dilated posterior urethra, vesicoureteral reflux, and valve leaflets can all be associated with posterior urethral valves and are visible on VCUG. Ureteropelvic junction obstruction and primary obstructive megaureter are both obstructions above the level of the urethra and are usually evaluated

with ultrasonography and a mercaptoacetyltriglycine (MAG3) renal scan or magnetic resonance urogram. Nephrolithiasis is typically evaluated using ultrasonography and computed tomography (CT) scan when necessary. Renal mass is initially evaluated by US, then MRI.

8. **a. Before discharge from the nursery.** Congenital adrenal hyperplasia may result in salt wasting, a medical emergency. Therefore neonates with ambiguous genitalia must be quickly evaluated and stabilized, prior to discharge home.
9. **c. The abdominal wall is normally strong, especially in infants with hydronephrosis.** Renal pathology accounts for approximately two-thirds of abdominal masses found in the neonate. Solid masses include neuroblastoma, congenital mesoblastic nephroma, teratoma, and renal enlargement due to renal venous thrombosis. The abdominal wall is normally weak in premature infants and on occasion in those with hydronephrosis.
10. **c. A presacral dimple less than 2.5 cm from the anal verge at birth may indicate spina bifida or cord tethering.** The lower back should be examined for any evidence of cutaneous markers of occult spinal dysraphisms that may account for abnormal bladder function. In a series of 207 neonates with sacral and presacral cutaneous stigmata, 40% of patients with atypical dimples were found to have occult spinal dysraphism. An “atypical” presacral dimple is defined as a dimple that is off center, more than 2.5 cm from the anal verge at birth, or deeper than 0.5 cm. Sacral hair tuft (hypertrichosis) may also be associated with spinal dysraphism.
11. **e. a, b, and c.** Although penile discharge and bruised vaginal mucosa can reflect sexual abuse, the possibility of sexual abuse should not be dismissed in the absence of physical examination findings. Only 11% of girls evaluated in a sexual abuse clinic demonstrated suggestive physical examination findings.
12. **d. After healing of the inflamed, denuded glans after circumcision.** Meatal stenosis is not unusual after circumcision. It may result from contraction of the meatus after healing of the inflamed, denuded glans tissue that occurs after retraction of the

foreskin or from damage to the frenular artery at the time of circumcision.

13. **d. Pure gonadal dysgenesis.** Particular attention to the symmetry of the examination is important if intersex/difference of sex development is thought to exist. A symmetric gonadal examination (gonads palpable on each side or impalpable on both sides) suggests a global disorder, such as congenital adrenal hyperplasia or androgen insensitivity. When a gonad is palpable, female congenital adrenal hyperplasia (ovaries are not palpable) and pure gonadal dysgenesis (bilateral streak gonads are not palpable) are ruled out.
14. **d. Urinary incontinence after a dry interval greater than 6 months.** Although urinary incontinence can be associated with infection, constipation, and neurologic disease, secondary urinary incontinence is defined as occurring after a dry interval greater than 6 months.
15. **e. b and c. A hydrocele that changes in volume suggests a patent processus vaginalis.** These infants are at risk for an inguinal hernia. The processus vaginalis is less likely to close after birth. If a hernia has been symptomatic, it should be corrected in the newborn period. A large scrotal hydrocele may still resorb and get smaller with time—distinction must be made from an abdominoscrotal hydrocele.

Chapter Review

1. The most common malignant abdominal tumor in infants is a neuroblastoma, followed by Wilms tumor.
2. Undescended testes are present in 30% of preterm neonates and 3% of full-term neonates; the testis will likely not descend after 6 months of age.
3. The most common prepubertal testicular/paratesticular tumor is teratoma, followed by rhabdomyosarcoma, epidermoid cyst, yolk sac tumor, and germ cell tumor, in that order.

4. Very few children hold urine and not stool. Conversely, children who retain stool nearly always retain urine.
5. Gross hematuria in the newborn is an emergency because it may indicate renal venous thrombosis or renal artery thrombosis.
6. In general, blunt renal trauma is treated nonoperatively, except when there is a major vascular avulsion or extensive urinary extravasation.
7. In the newborn, the foreskin is adherent to the glans, and adhesions should not be separated unless a circumcision is performed.
8. A positive dipstick for blood requires a microscopic examination. Absence of red blood cells in the microscopic examination indicates hemoglobinuria or myoglobinuria.
9. Continuous leakage of urine in a girl should suggest ectopic ureter.
10. Infants younger than 6 months of age and uncircumcised male infants are at increased risk for urinary tract infections.
11. A patients with a myelomeningocele is at increased risk for latex allergy.
12. Most microscopic hematuria in children is transient, and the source is not identified. The most commonly identified etiology of asymptomatic microhematuria in children is hypercalciuria.
13. Beckwith-Wiedemann syndrome is caused by a mutation on chromosome 11p15.5. Clinical features include macroglossia, nephromegaly, organomegaly (hepatosplenomegaly), macrosomia (gigantism), and hemihypertrophy. Many of the affected infants have hypoglycemia in the first few days of life. These individuals are at increased risk for adrenal carcinoma, Wilms tumor, hepatoblastoma, and rhabdomyosarcoma.
14. Renal pathology accounts for approximately two-thirds of abdominal masses found in the neonate.
15. If a difference of sex development (intersex) is thought to exist, a symmetric gonadal examination (gonads palpable on each side or impalpable on both sides) suggests a global disorder, such as congenital adrenal hyperplasia or androgen insensitivity. If the gonads are nonpalpable, congenital adrenal hyperplasia is likely.
16. A hydrocele that changes in volume suggests a patent processus vaginalis.

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Pediatric Urogenital Imaging

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Keywords

computerized tomography; contrast cystography; magnetic resonance imaging; nuclear renography; pediatric imaging; sonography

Questions

1. All of the following statements regarding prenatal sonography are true EXCEPT:
    a. amniotic fluid contains fetal urine before 10 weeks of age.
    b. amniotic fluid contains fetal urine by 10 to 12 weeks of age.
    c. urine fills the fetal bladder by 20 weeks of age.
    d. the fetal bladder never completely empties despite normal bladder cycling.
    e. in the fetal kidney, prominent corticomedullary differentiation can be confused for hydronephrosis.
2. All of the following statements about postnatal ultrasound findings of hydronephrosis are true EXCEPT:
    a. hydronephrosis may be obstructive or nonobstructive
    b. hydronephrosis is always an indication of obstruction.
    c. hydronephrosis can be more fully evaluated by functional imaging to identify obstruction.

d. hydronephrotic pelvicalyceal system can be differentiated from renal cysts based on whether the fluid spaces do not or do communicate.
e. autosomal recessive polycystic kidneys are characterized by homogeneously bright kidneys that are enlarged.

3. Identify the false statement about imaging for urinary tract infection.
    a. Sonography is very sensitive for the detection of vesicoureteral reflux.
    b. Sonography is not very sensitive for the detection of acute pyelonephritis.
    c. Sonography should be considered during acute infection when the patient fails to respond to antibiotic treatment.
    d. Contrast-enhanced sonography may soon supplant voiding cystourethrogram (VCUG) as a means to diagnose vesicoureteral reflux.
    e. Dimercaptosuccinic acid (DMSA) renal scan is highly accurate for the detection of acute pyelonephritis and renal scarring.
4. Identify the false statement about imaging the acute scrotum.
    a. Surgical exploration for torsion should not be delayed to obtain scrotal imaging if the history and physical examination suggests that diagnosis.
    b. Salvageability of the testicle based on sonographic appearance is difficult to predict.
    c. Epididymo-orchitis and spontaneous detorsion can *both* demonstrate hyperemia on sonography.
    d. Torsion of the appendix testis can be seen as an enlarged and avascular lower pole testicular nodule with surrounding hyperemia.
    e. Most prepubertal testicular masses are benign.
5. The four “key elements” to a successful diuresis renogram for the evaluation of hydronephrosis are:
    a. hydration, selection of the appropriate region of interest, bladder drainage as appropriate, timing of diuretic administration.

b. sedation, selection of the appropriate region of interest, bladder drainage as appropriate, timing of diuretic administration.
c. hydration, selection of the appropriate region of interest, bladder drainage as appropriate, timing of diuretic administration to occur exactly when furosemide is administered.
d. hydration, selection of the appropriate region of interest, catheterization of the bladder on completion of the test, timing of diuretic administration.
e. hydration, selection of the region of interest tightly around the renal cortex, bladder drainage as appropriate, timing of diuretic administration.

Answers

1. **a. Amniotic fluid contains fetal urine before 10 weeks of age.** Amniotic fluid does not contain fetal urine before 10 weeks of age.
2. **b. Hydronephrosis is always an indication of obstruction.** Hydronephrosis is not always an indication of obstruction.
3. **a. Sonography is very sensitive for the detection of vesicoureteral reflux.** Sonography is not sensitive for the detection of vesicoureteral reflux.
4. **d. Torsion of the appendix testis can be seen as an enlarged and avascular lower pole testicular nodule with surrounding hyperemia.** Torsion of the appendix testis is not seen as an enlarged and avascular lower pole testicular nodule with surrounding hyperemia.
5. **a. Hydration, selection of the appropriate region of interest, bladder drainage as appropriate, timing of diuretic administration.** The four key elements to a successful diuresis renogram are hydration, selection of the appropriate region of interest, bladder drainage as appropriate, and timing of diuretic administration.

Chapter Review

1. Sonography is the initial examination to perform for suspect obstruction, as well as being helpful for longitudinal follow-up.
2. Based upon the sonographic findings, ancillary imaging (such as voiding cystourethrography and/or diuresis renography) might be considered.
3. A normal infant renal sonogram can be confused with mature hydronephrosis because of hypoechoic renal pyramids with a distinct corticomedullary junction.
4. Diagnosis of obstruction in children with diuresis renography requires review of all parameters of the test because variations in technique can skew results considerably.
5. Sonographically, multicystic dysplastic kidney (MCDK) is characterized by echolucent spaces (cysts) of varying sizes that are randomly distributed throughout the kidney with a paucity of renal parenchyma that is hyperechoic.
6. Autosomal recessive polycystic kidney disease (ARPKD) typically affects both kidneys and is characterized by enlarged and echogenic parenchyma.
7. Autosomal dominant polycystic kidney disease (ADPKD) typically affects both kidneys and is characterized by replacement of the renal parenchyma by cysts.
8. Although patient age and hallmark sonographic features suggest the diagnosis of renal masses, contrast-enhanced computed tomography (CT) scan of the abdomen should follow.
9. Numerous guidelines have been published on the radiographic evaluation following pediatric urinary tract infection, and sonography is used as an initial screening modality despite its low yield.
10. A normal sonogram is insufficient to risk-stratify a child with a febrile urinary tract infection (UTI) and is not a good predictor for vesicoureteral reflux (VUR).
11. The “bottom-up” approach relies on VCUG—following US—to identify all cases of VUR that may result in overtreatment of low-

grade VUR.

12. The “top-down” approach replaces screening US with a DMSA renal scan to identify acute pyelonephritis and/or renal scarring. Proponents of this algorithm recommend VCUG only in response to abnormal DMSA scans as an indicator of high-risk patients.
13. Abdominal sonography is useful as part of the acute trauma evaluation and compares favorably to contrast-enhanced CT scans. Sonography should be considered for the longitudinal evaluation of known renal injuries, particularly to assess resolution of perinephric hematomas or urinomas.
14. Ionizing radiation is used to evaluate genitourinary trauma, frequently with the addition of intravenous or intraluminal iodinated contrast.
15. Radiographic evaluation is incomplete unless delayed views of the renal parenchyma and collecting system are obtained on CT scan, or lateral projections of the bladder, bladder neck, and urethra.
16. In addition to careful history and physical examination, scrotal sonography provides much of the information needed to discriminate between inflammation, neoplasm, trauma, and testicular torsion.
17. Nuclear scintigraphy may be used to evaluate testicular torsion but provides little additional information over what can be gleaned from sonography.
18. Sonography for the undescended testis is unreliable and should not be used, perhaps with the exception of seeking the inguinal undescended testis in an overweight boy.